

ADITI GUPTA

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SUMMARY

AI/ML undergraduate with research experience at TCS Performance Research Lab and FOSSEE, IIT Bombay. Skilled in machine learning, deep learning, generative AI, and computer vision, with experience building end-to-end AI systems and optimizing cloud deployments. Awarded a research honorarium by IIT Bombay for contributions to predictive modeling and applied AI research.

EDUCATION

Vellore Institute of Technology Bhopal University

B.Tech in CSE (Specialization in AI & ML) | CGPA: 8.9/10

Bhopal, India

Aug 2023 – Present

DAV Public School

Senior Secondary Education (Grade XII), CBSE – 93.2%

Gurugram, India

Apr 2022 – Mar 2023

SKILLS

Programming Languages: Python, C++, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Transformers, LLM

Frameworks & Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, Keras, MediaPipe, OpenCV

Tools & Platforms: Flask, Streamlit, AWS, Render, Git, GitHub, Jupyter Notebook, VS Code, DWSIM, Google Vertex AI

EXPERIENCE

TCS, Performance Research Lab | Research & Innovation Summer Intern | Mumbai

May 2026 – July 2026

Systems Optimization in Generative AI Based Applications

- Built a cloud accelerator recommendation and LLM inference prediction system using XGBoost to estimate latency and throughput across 86 model hardware configurations.
- Engineered automated hardware resolution logic with multi-level fallback and nearest-parameter model substitution, reducing manual configuration overhead for LLM deployment planning.

FOSSEE, IIT Bombay | Machine Learning Intern | Remote

Oct 2025 – Feb 2026

Crude Oil Characterization & Product Properties Estimation | [GitHub](#) | [Certificate](#) | [Published Work \(FOSSEE\)](#)

Python, XGBoost, Scikit-Learn, TensorFlow, Keras

- Built a data extraction and validation pipeline for 110+ global crude oil assay files with domain-specific checks across hydrocarbon composition, kinematic viscosity, and quality indices.
- Benchmarked 12+ ML algorithms, achieving R^2 up to 0.91 (XGBoost) for kinematic viscosity and 0.73 (SVR) for hydrocarbon composition; work published on the FOSSEE website; awarded honorarium by FOSSEE, IIT Bombay for contribution to the project.

PROJECTS

Real-Time Indian Sign Language Fingerspelling Translator | [GitHub](#)

Apr 2026

Python, Scikit-Learn (MLPClassifier), MediaPipe, OpenCV

- Built a real-time Indian Sign Language fingerspelling translator recognizing all 26 ISL alphabet gestures (A–Z) with hold-to-confirm word-building logic at 30+ FPS on standard consumer hardware.
- Engineered a 63-dimensional landmark-based feature pipeline using 21 MediaPipe hand points and trained a 3-layer MLPClassifier (256→128→64) on 5,277 samples from 7,800 ISLRTC images, achieving 96.69% test accuracy.

Binary Distillation Surrogate Model | [GitHub](#)

Sep 2025

Python, XGBoost, Random Forest, DWSIM, Scikit-Learn

- Developed ML surrogate models replacing expensive DWSIM simulations to predict distillate purity (xD) and reboiler duty (QR); XGBoost achieved R^2 of 0.95 and 0.93 respectively.
- Generated a 373-sample dataset via DWSIM simulations varying 6 operating parameters and benchmarked 6 regression models using RandomizedSearchCV; project secured selection for the FOSSEE IIT Bombay internship.

Student Performance Predictor | [GitHub](#) | [Live Demo](#)

Sep 2025

Python, CatBoost, XGBoost, Flask, Render

- Built an end-to-end ML pipeline and automated preprocessing with Scikit-Learn ColumnTransformer (StandardScaler + OneHotEncoder) across 1,000 records and 8 features.
- Evaluated 10+ regression algorithms with GridSearchCV; Ridge Regression achieved R^2 of 0.88. Deployed as a production Flask web app on Render with a responsive UI for real-time predictions.

CERTIFICATIONS

- [Applied Machine Learning in Python](#) – University of Michigan
- [Internet of Things Certification \(Elite + Gold\)](#) – NPTEL | Top 5% National
- [Cloud Computing Certification \(Elite\)](#) – NPTEL

ACHIEVEMENTS

Awarded Research Honorarium by FOSSEE, IIT Bombay | Served as Head Girl and School Captain at Delhi Public School